The Corridor (Valley / Highway / Chokepoints & Convoy Warfare)

Executive Summary

The Corridor is a **valley+highway battlefield** built for **convoy warfare. chokepoint control. lone+wave ambushes. and logistics disruption**. The terrain features a narrow valley floor. elevated highway seements. tunnels. overpasses. fuel depots. roadside industrial zones. and steep rock walls that funnel movement into predictable lanes. The map emphasizes Obsidian Protocol's autonomous convoy logic. sensor-driven ambush detection. comms-relay networks. and logistics under extreme linear constraints. In AR thi real Light/Art. the map anchors at ~4–5 m viewing distance. covering a 10×10 m physical area representing ~1 km² of valley terrain. Key features include multi-tier highway combat. tunnel chokepoints. ambush-ready rock shelves. and strategic control of convoy routes.

Design Goals & Gameplay

Convoy Warfare & Route Control
The Corridor prioritizes convoy escort, interception, and disruption. Highway segments create predictable movement paths ideal for ambushes and defensive setups.
Chokepoint+Driven Combat
Tunnels, overpasses, and narrow valley bends create natural chokepoints. Controlling these determines the flow of the entire battle.
Autonomy & Route-Selection Logic
Autonomous AI chooses between upper highway lanes, lower valley roads, tunnel bypasses, or off-road rock paths. Units adapt to blocked roads and ambush threats.
Sensor & Information Warfare
Dust, exhaust, tunnel darkness, and rock occlusion distort sensors. Relay towers on cliffs restore clarity. Destroying relays creates blind zones along the highway.
Logistics & Resources
Fuel depots, roadside industrial yards, and highway checkpoints serve as resupply nodes. Convoys must navigate exposed stretches and avoid ambush-prone bends.
Deployment & Progress
A fixed deployment budget (~10,000 points) forces players to choose between convoy escorts, heavy armor, or ambush-specialized units. Blue deploys from the southwest valley checkpoint; Red deploys from the northeast elevated highway command.

Spatial Constraints (Nreal AR)

Device Capabilities
Nreal Light/Air: ~52° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.
Playable Area & Scaling
10×10 m physical area = 1 km² valley zone (1:100 scale). Designed for tabletop AR.
Tracking & Occlusion
Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System
1000×1000 grid (1 unit = 1 m). Origin (0,0) = southwest corner.
Key Locations
Blue HQ (Valley Checkpoint) — (0,0). 140×140 m Roadblock. watchtower. convoy staging area.
Red HQ (Highway Command Overlook) — (1000,1000). 140×140 m Elevated command post with radar mast and AA emplacements.
Central Highway Spine — (500,450). 300×200 m Main combat zone: multi-lane elevated highway + lower valley road.
Fuel & Logistics Depot — (250,800). 180×180 m Fuel tanks. tents. trailers. cargo crates.
Tunnel Complex — (800,300). 150×150 m Two tunnels. ventilation shafts. interior ambush routes.
Valley Bend — (1,450 → 400,900) Narrow rock corridor: extreme chokepoint.
Lower Road Route — (200,0 → 450,350) Safer but slower convoy path with partial cover.
Cliff Shelf A — (650,150). +240 m Ambush vantage point overlooking highway.
Cliff Shelf B — (150,650). +260 m Comms relay tower and recon hub.
Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

Valley Bend
The most critical chokepoint; perfect for ambushes and artillery traps.
Tunnel Complex
Interior routes allow stealth flanking and sudden close-range engagements.
Highway Spine
Long-range sightlines; ideal for convoy escort or disruption.
Lower Road Route
Safer but predictable; vulnerable to cliffside overwatch.
Sightlines
Cliff Shelf A provides long-range sensor coverage (~350 m). Dust and tunnel darkness create fog-of-war pockets.

Deployment Zones

Blue Deployment: Qal valley checkpoint + convoy routes.
Red Deployment: NE highway command overlook + AA emplacements.
Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain
Valley floor, cliffs, rock shelves, elevated highway, tunnels.
Vegetation
Sparse shrubs, dry grass patches.
Buildings & Props
Military: Watchtowers, bunkers, radar dishes, AA guns. Industrial: Fuel tanks, pipelines, generators, cargo yards. Logistics: Trucks, crates, fuel drums, convoy vehicles.
Effects
Dust clouds, exhaust haze, tunnel fog, wind gusts.
Audio
Engine noise, wind, distant artillery, radio chatter.

LOD & Budget

Asset Cate	Count	Tri/Item	Total Tris
Cliffs/Roc	3	200k	600k
Highway/	4	80k	320k
Buildings/	15	50k	750k
Roads/Pa	4	30k	120k
Vegetatio	100	2k	200k
Props	50	10k	500k
Total	—	—	**~2.49M**

Texture Budget
2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins
Unreal ARTemplate + Nreal SDK (NREAL). ARKit/ARCore enabled.
Rendering
Forward renderer, stationary lights, baked GI, aggressive culling.
Anchors
Single spatial anchor for entire map.
Interaction
Phone pointer for pan/zoom; HUD as screen-space widgets.
Multiplayer
Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones
1. Prototype Layout — Month 1 Block-out highway. tunnels. valley bend. AR anchor test.
2. Core Systems — Month 2 NavMesh. convoy logic. ambush detection. autonomy behaviors.
3. AR Integration — Month 3 Anchors. plane detection. basic interactivity.
4. Content Dr — Month 4–5 Final models. textures. lighting. audio.
5. Gameplay & Polish — Month 6–7 Objectives. balance. VFX/dust. exhaust hazel.
6. Testing & QA — Month 8 Multiplayer stress test. sensor validation.

Testing Plan

Autonomy
Verify convoy logic, ambush detection, tunnel navigation.
Comms/Sensors
Test relay destruction, fog-of-war updates, LOS checks.
Performance
≥60 Hz AR render on target device.
QA Checklist
Navigation, chokepoint behavior, convoy pathing, deployment budget enforcement, anchor drift, sensor accuracy.